

Qwen Daemon Stack Integration & RTX 5090 Production Deployment

Project proposal and execution plan to finish the pre-model backend and deploy Phase 2 (GPU + Qwen models).

Prepared for Rico (DataismLab)	Prepared by Varun Chaturvedi	Date October 7, 2026
Version 1.0.0	Target RTX 5090 (32GB VRAM), Hyper-V Ubuntu VM, Qwen- native agentic stack	

1. Executive summary

This proposal covers the engineering plan, architecture, schedule and charges to finalize the pre-model backend and complete the Phase 2 GPU + Qwen Models deployment for the Qwen Code / Daemon system.

The AI-Lab Ubuntu VM moves from CPU-only to a production-hardened, GPU-accelerated agentic coding workspace on an RTX 5090 32GB. The work follows a Qwen-centered approach: native Qwen model weights and established vendor inference backends (vLLM, NVIDIA Container Toolkit), while preserving every existing working component.

2. Technical execution roadmap





Step 0: Pre-model backend audit and completion

Validate and seal the four pre-model layers without rewriting working components.

1. **0.1 Daemon service and harness validation.** Inspect systemd/supervisor units, IPC sockets and environment configuration for the Qwen Code / Daemon runner.
2. **0.2 Foreman / Worker delegation harness.** Confirm Foreman cleanly dispatches task breakdowns to specialized sub-agents over the Agent Communication Protocol (ACP).
3. **0.3 Compliance and permission interception.** Verify hooks that intercept destructive bash/filesystem actions (`rm -rf` , file overwrites, unapproved outbound requests) before execution.
4. **0.4 Tooling, MCP and memory / RAG baseline.** Validate MCP server connectivity, vector store / SQLite storage and context retrieval without GPU dependency.

Milestone 1: RTX 5090 passthrough and NVIDIA container stack

Expose the physical 32GB RTX 5090 from the Windows host to the Ubuntu VM and container runtime.

Hyper-V GPU passthrough (DDA). The RTX 5090 needs native PCIe exposure for direct CUDA access (`/dev/nvidia*` , `nvidia-smi`). Host actions, in an admin PowerShell:

1. Identify the PCI location path (`Get-PnpDevice -Class Display`).
2. Disable and dismount the device from the host: `Dismount-VMHostAssignableDevice` .
3. Configure VM memory and MMIO space, then assign the device:

```
Set-VM -VMName "AI-Lab" -AutomaticStopAction TurnOff
Set-VM -VMName "AI-Lab" -GuestControlledCacheTypes $true -LowMemoryMappedIoSpace 3Gb -HighMemoryMappe
Add-VMAssignableDevice -VMName "AI-Lab" -LocationPath $pciLocationPath
```

Fallback: GPU-PV with the WSL2/dxgkrnl bridge if the Windows host edition or topology blocks bare-metal DDA.

Ubuntu guest configuration. Install the NVIDIA production driver (570.x+, Blackwell support), CUDA Toolkit 12.8+ and `nvidia-container-toolkit` . Configure `/etc/docker/daemon.json` with `default-runtime: nvidia` and isolated rootless cgroups.

Verification. Run `nvidia-smi` natively, then a non-privileged container test:

```
docker run --rm --gpus all nvidia/cuda:12.8.0-base-ubuntu24.04 nvidia-smi
```

Milestone 2: Qwen local inference engine and router

A secure, OpenAI-compatible local inference gateway isolated from host vulnerabilities.

- **Backend:** vLLM in a dedicated non-privileged Docker container. The Docker socket (`/var/run/docker.sock`) is never mounted. An isolated internal bridge network exposes only the inference port (`http://127.0.0.1:8000/v1`). PagedAttention and FP8/AWQ tensor cores are enabled for Blackwell.
- **Router:** a lightweight local gateway (or vLLM multi-model routing) so Qwen Code can direct prompts to designated model roles.

Milestone 3: Qwen model sizing and VRAM optimization (32GB budget)

Maximize coding capability and multi-role efficiency without exceeding 32GB VRAM.

Role	Model	Quantization	Model VRAM	KV cache (32k ctx)	Total	Mod
Coding specialist	Qwen2.5-Coder-32B-Instruct	AWQ / FP8	~19.5 GB	~8.5 GB (FP8 KV)	~28.0 GB	Prim resic
General / architecture	Qwen2.5-72B-Instruct	AWQ / GPTQ	~9.2 GB	~4.5 GB	~13.7 GB	Dyni / Tie resic
Worker / utility / compliance	Qwen2.5-Coder-72B-Instruct or Qwen2.5-72B	AWQ / INT4	~4.8 GB	~2.5 GB	~7.3 GB	Co- resic with 14B, fast sub- ager

- **Mode 1, max performance coding:** Qwen2.5-Coder-32B (AWQ) stays resident (~28GB). A quantized 3B/7B utility model uses residual VRAM or dynamic swap for fast JSON/compliance validation.
- **Mode 2, high-concurrency agent swarm:** Qwen2.5-Coder-14B (13GB) and Qwen2.5-7B

Milestone 4: Complete stack integration

- Connect Qwen Code through Qwen Serve (ACP) to `http://127.0.0.1:8000/v1`.
- Wire the Foreman Agent to dispatch parallel implementation tasks to Worker Agents.
- Enforce the compliance layer: system prompts and deterministic filters block destructive bash commands (`rm -rf`, raw disk writes, unpermitted external calls) pending explicit human confirmation.
- Connect MCP servers (filesystem, git, terminal) in a sandboxed, restricted execution directory.
- Verify memory / RAG: index project files, test vector embeddings and contextual recall during prompt assembly.

Milestone 5: Production acceptance benchmark and audit report

A full validation suite and a signed-off final report.

1. Normal chat and prompt fidelity: multi-turn dialogue, context retention.
2. Coding and multi-file refactoring: syntax generation, cross-file context.
3. Foreman / Worker delegation: hierarchical breakdown and resolution.
4. Compliance and approval enforcement: dangerous commands verified blocked.
5. MCP tools: successful invocation and structured response parsing.
6. Persistent memory / RAG: document retrieval and cross-session lookup.
7. Sandbox isolation: confinement, no escape, non-root privileges.
8. GPU inference benchmarks: TTFT under 400ms; tokens/sec at 4k, 16k and 32k context; VRAM, temperature and power stability; failure recovery and daemon self-healing after simulated crashes.

3. Timeline and schedule

An accelerated two-day sprint.

Phase / milestone	Hours	Schedule	Deliverables
Phase 1: Pre-model backend audit and completion	4	Day 1 morning	Verification of 4 pre-model steps, daemon config validation, non-destructive fix list
Phase 2.1: Hyper-V RTX 5090 passthrough and	6	Day 1 afternoon	DDA configured, NVIDIA 570+ and Container Toolkit operational, Nvidia smi passing

GPU passthrough and drivers			
Phase 2.2 and 2.3: Inference engine and model deployment	5	Day 1 evening	vLLM/router container deployed, Qwen2.5 weights downloaded/quantized, 32GB VRAM tuned
Phase 2.4: End-to-end integration and hardening	8	Day 2 morning	Qwen Code, ACP, Foreman, Compliance, MCP, RAG, Sandbox connected
Phase 2.5: Acceptance benchmark and audit report	5	Day 2 afternoon	Benchmark suite, stress tests, PASS/FAIL report and documentation
Handover, review and sign-off	2	Day 2 evening	Live walkthrough with Rico, review, production sign-off
Total	30	2 days	Production-ready Qwen Daemon system

4. Time and charges

One-time flat project fee (turnkey delivery)

All-inclusive fixed price covering the entire scope, from the pre-model backend to the final production benchmark report.

Item	Detail
Total one-time fee	\$300 USD (₹26,000 INR)
Estimated effort	30 hours total
Turnaround	2 days (intensive sprint)

Scope covered

1. Audit and completion of the 4 pre-model backend components (non-destructive).
2. Hyper-V RTX 5090 (32GB) DDA PCIe passthrough, NVIDIA 570+ drivers and Container Toolkit.
3. Isolated local inference container (vLLM) with Qwen2.5-Coder-32B (AWQ/FP8) and worker models.
4. Full stack binding: Qwen Code, ACP, Foreman/Workers, Compliance, MCP, RAG,

Sandbox.

5. Production acceptance benchmark suite and formal PASS/FAIL audit report.

Payment terms

- 50% advance at kickoff: \$150 / ₹13,000
- 50% on delivery and acceptance of the production benchmark report: \$150 / ₹13,000
- Or a single invoice on delivery, if preferred.

Alternative: time and materials

Rate: \$10/hour (₹870/hour). Estimated 30 hours, about \$300 (₹26,000), billed against verified timesheets and task logs.

Included

- **Zero-disruption guarantee:** existing working configurations are preserved without rebuilds.
- **Isolated security architecture:** no unprivileged host access, no exposed Docker sockets.
- **14-day post-delivery warranty:** free fixes and configuration adjustments for issues arising from the delivered scope.
- **Documentation:** full configuration docs, recovery SOPs and model-swap scripts.

Client prerequisites

1. Administrator access to the Windows host (or an assisted session) to run the Hyper-V device assignment commands (`Add-VMAssignableDevice`).
2. Stable internet bandwidth on the host/VM to download Qwen model weights (about 20–30GB total).

5. Client message draft for Rico

Hi Rico,

Thanks for providing the SSH credentials and detailed specification.

I have thoroughly reviewed the existing environment and prepared the execution plan to finish the pre-model backend and deliver Phase 2 (RTX 5090 hookup, Qwen local inference, model sizing for 32GB VRAM, complete stack integration, and the production benchmark report).

Here is the summary of the plan, timeline, and commercial terms:

1. Scope highlights:

- Non-destructive: preserve and finalize the 4 pre-model backend components without rebuilding what is already working.
- GPU integration: Hyper-V DDA / PCIe passthrough so Ubuntu has direct access to the physical RTX 5090, followed by NVIDIA 570+ driver and Container Toolkit.
- Qwen inference stack: vLLM in an isolated, unprivileged container (no Docker socket exposure), hosting Qwen2.5-Coder-32B (AWQ/FP8) plus auxiliary models tailored to the 32GB VRAM budget.
- Full stack integration: Qwen Code -> Qwen Serve/ACP -> Foreman/Workers -> Compliance/Guardrails -> MCP tools -> Memory/RAG -> Sandbox -> Local GPU.
- Deliverable: a Production Acceptance Report with PASS/FAIL matrices across all layers, latency/throughput metrics, VRAM stats and production sign-off.

2. Timeline and effort:

- Estimated turnaround: 2 days (intensive sprint of ~30 engineering hours total).

3. Charges:

- One-time fixed project fee: \$300 USD flat (approx. ₹26,000 INR, based on 30 hours @ \$10/hr). Single turnkey engagement with no milestone splits.
- Payment: 50% deposit at kickoff (\$150) and 50% on final delivery and benchmark sign-off (\$150).
- Includes a 14-day warranty for post-launch stabilization and bug fixes.
(Alternatively, available hourly at \$10/hr.)

Please let me know if this plan aligns with your schedule and expectations so we can kick off Step 1 right away.

Best regards,
Varun

